JAMES RICHARDSON

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PROFESSIONAL SUMMARY

Data scientist with 7 years of experience building machine learning systems that solve real business problems. I've spent the last 4 years at senior level working across financial services and tech, focusing on predictive modelling and data infrastructure. I'm most comfortable working end-to-end from data exploration through to production deployment, and I've mentored 5 junior data scientists throughout my career. I'm based in Manchester and prefer roles where I can contribute to both technical decisions and team strategy.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, Java

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM

Data Processing: Pandas, NumPy, PySpark, Airflow

Databases: PostgreSQL, MySQL, MongoDB, Redshift

Cloud Platforms: AWS (EC2, S3, SageMaker), Azure ML

Visualisation: Matplotlib, Seaborn, Tableau, Plotly

Development: Git, Docker, Linux, Jupyter, VS Code

Statistical Analysis: Hypothesis testing, A/B testing, time series analysis, Bayesian methods

WORK EXPERIENCE

Senior Data Scientist

Quantum Insights Ltd

quantuminsights.io

Manchester, United Kingdom

March 2021 - Present

Led development of a customer churn prediction model for 2 million+ users, reducing customer loss by 15% and generating 3.2 million pounds annual revenue impact. Designed and implemented a real-time recommendation engine using collaborative filtering and deep learning, increasing product engagement by 28%. Built automated data pipelines using Airflow and PySpark to process 500GB daily transactions, reducing reporting latency from 6 hours to 15 minutes. Mentored 3 junior data scientists and established coding standards and model validation protocols across the team.

Data Scientist

Trinity Financial Services plc

www.trinityfinancial.co.uk

Manchester, United Kingdom

July 2018 - February 2021

Developed credit risk models using logistic regression and gradient boosting techniques, achieving 89% AUC and reducing loan defaults by 12%. Created a fraud detection system leveraging ensemble methods that flagged 94%

of fraudulent transactions with minimal false positives, saving the company 2.1 million pounds annually. Built dashboards and automated reporting using SQL and Tableau, serving insights to 40+ stakeholders across risk and compliance departments. Conducted feature engineering on 300+ variables from transaction data, improving model performance through domain knowledge and statistical testing.

Junior Data Scientist

Vertex Analytics Solutions

vertexanalytics.net

Manchester, United Kingdom

November 2016 - June 2018

Developed predictive models for customer lifetime value using regression and tree-based methods, informing 5 million pounds marketing budget allocation. Extracted and cleaned data from multiple sources using Python and SQL, working with datasets containing 50+ million records. Performed exploratory data analysis on marketing campaign data and presented findings to non-technical stakeholders through visualisations and reports. Participated in code reviews and contributed to internal Python packages used by 12 analysts across the company.

EDUCATION

MSc Data Science

University of Manchester

Manchester, United Kingdom

2014 - 2016

BSc Mathematics

University of Nottingham

Nottingham, United Kingdom

2011 - 2014

GITHUB REPOSITORIES

github.com/jrichardson-data/churn-prediction-ensemble

Churn prediction model comparing 7 different algorithms with cross-validation and hyperparameter tuning using Bayesian optimisation.

github.com/jrichardson-data/credit-risk-framework

Python package for credit risk model development including feature engineering, model selection, and validation utilities.

github.com/jrichardson-data/timeseries-anomaly-detection

Unsupervised anomaly detection in time series data using isolation forests and autoencoder neural networks.

github.com/jrichardson-data/nlp-sentiment-pipeline

End-to-end sentiment analysis pipeline using transformers and BERT for social media monitoring.